# MIT Assignment-08

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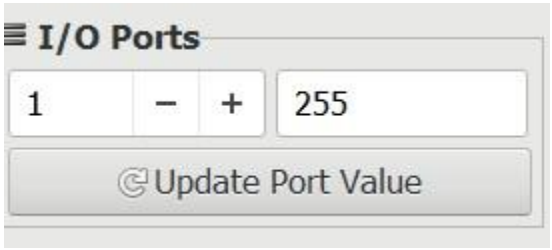
1

Write a Program to count continuously in hexadecimal from FFH to 00H in a system with 0.5µs cock period. Use register C to set up a one milisecond delay between each count and display the numbers at one of the output port. MVI A, 00H

```
LOOP: OUT 01H

      CALL DELAY

      DCR A
      CPI 0FFH
      JNZ LOOP

      OUT 01H

      HLT

DELAY: MVI C, 00H
D1:    NOP
       NOP
       DCR C
       JNZ D1
       RET
```



| | |
|---|---|
| 2 | Write a program to count from 0 to 9 with a one second delay between each count. At the count of 9,the counter should reset itself to 0 and repeat sequence continuously. Use register pair HL to set up the delay,and display each count at one of the output port. Assume the clock frequency of the microprocessor is 1 MHz. |

```
; Program to count 0-9 with 1 second delay
; Clock Frequency: 1 MHz
; Port: 01H (Assumed)

        ORG 0000H
START:  MVI C, 00H      ; Initialize register C as the counter (0)

DISPLAY:MOV A, C        ; Move the current count to Accumulator
OUT 01H         ; Output the count to Port 01H

        ; --- 1 Second Delay Subroutine using HL Pair ---
        LXI H, A2C2H    ; Load HL with A2C2H (calculated for 1s)
DELAY:  DCX H           ; Decrement HL pair (6 T-states)
        MOV A, L        ; Move L to A (4 T-states)
        ORA H           ; OR with H to check if HL = 0 (4 T-states)
        JNZ DELAY       ; If not zero, jump back to DELAY (10 T-states)
        ; ----------------------------------------------

        INR C           ; Increment the counter
        CPI 0AH         ; Compare Accumulator with 10 (0AH)
        MOV A, C        ; Move C to A for comparison
        CPI 0AH         ; Check if count has reached 10
        JZ START        ; If count is 10, reset to 0
        JMP DISPLAY     ; Otherwise, display the next number
```

3

Calculate the delay in the following loop, assuming the system clock period is 0.33 μs:

| Label | Mnemonics | 8085 T-states |
|---|---|---|
| | LXI B,12FFH | 10 |
| DELAY: | DCX B | 6 |
| | XTHL | 16 |
| | XTHL | 16 |
| | NOP | 4 |
| | NOP | 4 |
| | MOV A,C | 4 |
| | ORA B | 4 |
| | JNZ DELAY | 10/7 |

Given:
Clock period = 0.33 μs
LXI B,12FFH → 12FFH = 4863 (loop count)

T-states per iteration:
DCX B (6) + XTHL (16) + XTHL (16) + NOP (4) + NOP (4) + MOV A,C (4) + ORA B (4) + JNZ (10) = 64 T-states

Last iteration:
JNZ = 7 instead of 10 → 64 - 3 = 61 T-states

Total T-states:
= (4862 × 64) + 61 + 10
= 311168 + 61 + 10
= 311239 T-states

Total delay:
= 311239 × 0.33 μs
≈ 102709 μs
≈ 102.7 ms

Final Answer: 102.7 ms

4

```
LOOP:  LXI   B,2384H
       DCX   B
       MOV   A,C
       ORA   B
       JNZ   LOOP
```

Given:
Clock period = 0.33 μs
LXI B,2384H → 2384H = 9092 (loop count)

T-states:
LXI B = 10
DCX B = 6
MOV A,C = 4
ORA B = 4
JNZ = 10 (jump), 7 (last)

T-states per iteration:
= 6 + 4 + 4 + 10
= 24 T-states

Last iteration:
= 24 - 3 = 21 T-states

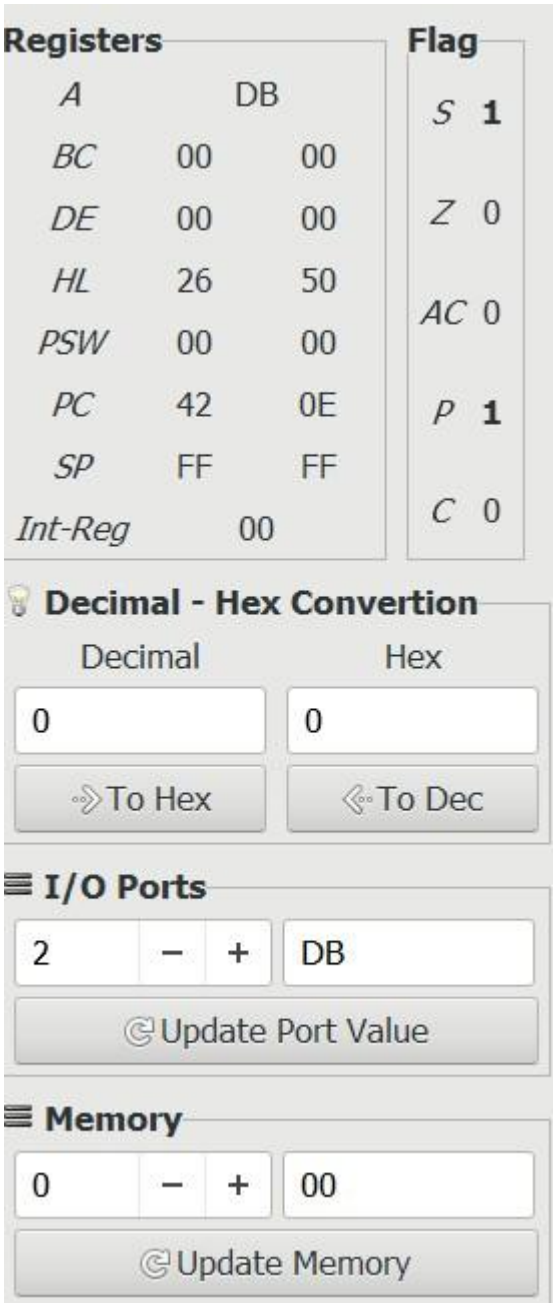
Total T-states:
= (9091 × 24) + 21 + 10
= 218184 + 21 + 10
= 218215 T-states

Total delay:
= 218215 × 0.33 μs
≈ 72010 μs
≈ 72.0 ms

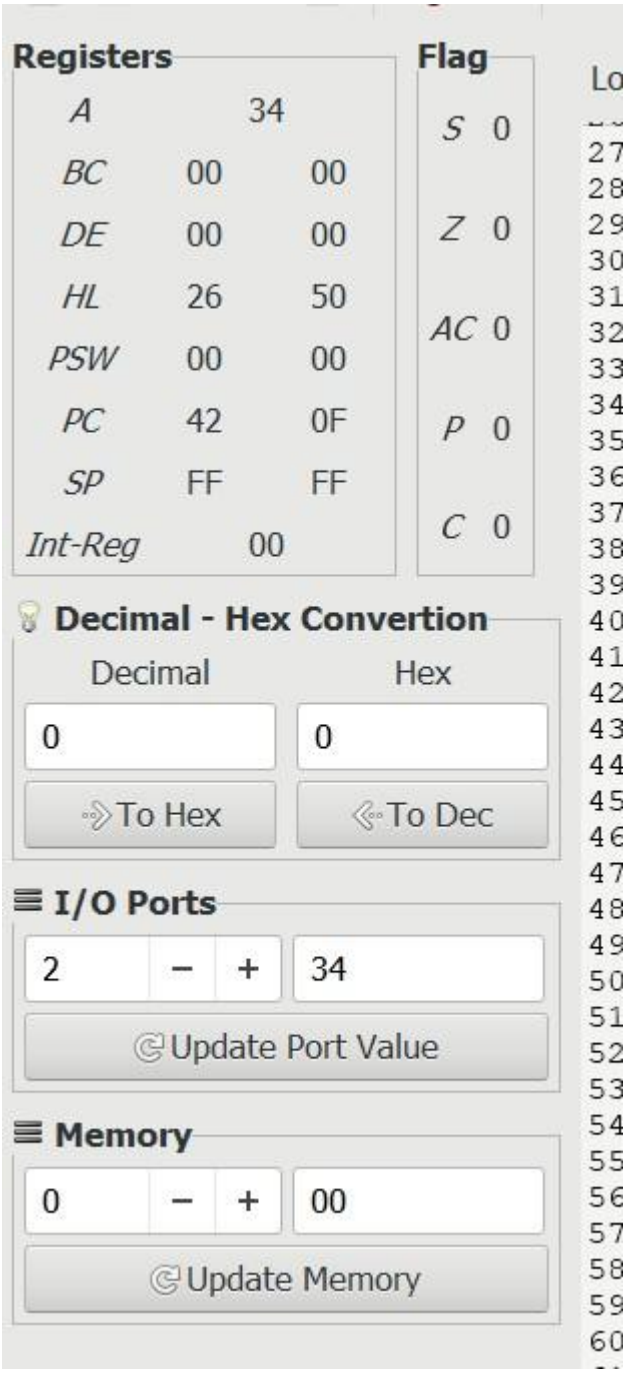
Final Answer: 72.0 ms

| | |
|---|---|
| 5 | A number is stored in memory location XX50H .Write a program to<br>1. Transfer the number from memory locations to A 2.<br>Store a result at output port 02H<br><br>LXI H,2650H<br>MOV A,M |
| | OUT 02H<br>HLT<br> <br> |

| | |
|---|---|
| | Write a subroutine to<br>1. To find 2's complement of a number stored in A<br>2. Return<br>COMP: CMA<br>INR A<br>RET |


Registers
A DB
BC 00 00
DE 00 00
HL 26 50
PSW 00 00
PC 42 0E
SP FF FF
Int-Reg 00
Flag
S 1
Z 0
AC 0
P 1
C 0
Decimal - Hex Convertion
Decimal
Hex
0
0
To Hex
To Dec
I/O Ports
2
DB
Update Port Value
Memory
0
00
Update Memory

| | |
|---|---|
| 6 | . Write a program to<br>1. Take number from input pot 1 and Transfer the numbers to A<br>2. Take another a number from memory location 2000H. 3. Store a result at output port 02H<br><br>MVI A,34H<br>STA 2000H<br><br>IN 01H<br>MOV B,A<br><br>LDA 2000H<br>ADD B<br><br>OUT 02H<br>HLT |

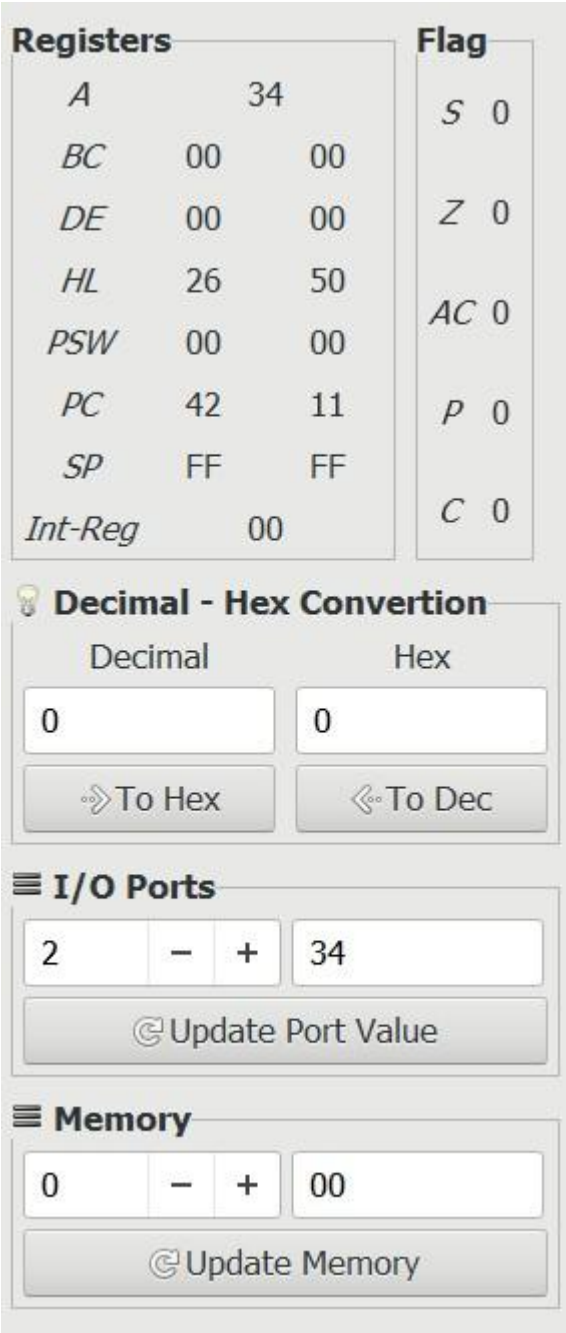

Registers
A 34
BC 00 00
DE 00 00
HL 26 50
PSW 00 00
PC 42 0F
SP FF FF
Int-Reg 00
Flag
S 0
Z 0
AC 0
P 0
C 0
Decimal - Hex Convertion
Decimal
Hex
0
0
To Hex
To Dec
I/O Ports
2
34
Update Port Value
Memory
0
00
Update Memory

Write a subroutine to 1.

Add both numbers

2. Return .

```
MVI A,34H
STA 2000H

IN 01H
MOV B,A

LDA 2000H

CALL ADDR

OUT 02H
HLT

ADDR: ADD B
      RET
```


Registers
A 34
BC 00 00
DE 00 00
HL 26 50
PSW 00 00
PC 42 11
SP FF FF
Int-Reg 00
Flag
S 0
Z 0
AC 0
P 0
C 0
Decimal - Hex Convertion
Decimal
Hex
0
0
To Hex
To Dec
I/O Ports
2
34
Update Port Value
Memory
0
00
Update Memory

7

| Line No. | Mnemonics |
|---|---|
| 1 | LXI SP,0400H |
| 2 | LXI B,2055H |
| 3 | LXI H,22FFH |
| 4 | LXI D,2090H |
| 5 | PUSH H |
| 6 | PUSH B |
| 7 | MOV A,L |
| ↓ | ↓ |
| 20 | POP H |

a. What is stored in the stack pointer register after the execution of line 1?

b. What is the memory location of the stack where the first data byte will be stored?

c. What is stored in memory location 03FEH when line 5 (PUSH H) is executed?

d. After the execution of line 6 (PUSH B), what is the address in the stack pointer register, and what is stored in stack memory location 03FDH?

e. Specify the contents of register pair HL after the execution of line 20 (POP H).

**(a) SP after line 1**

SP = 0400H

**(b) First data byte stored location**

- PUSH decreases SP first

SP → 03FFH (first location used)

**(c) Value at 03FEH after PUSH H** H = 22H, L = FFH Execution:

- SP → 03FF → store H (22H)
- SP → 03FE → store L (FFH)

Memory[03FEH] = FFH

**(d) After PUSH B** B = 20H, C = 55H Current SP = 03FEH

Execution:

- SP → 03FD → store B (20H)
- SP → 03FCH → store C (55H)

SP = 03FCH

Memory[03FDH] = 20H

**(e) After POP H (line 20)** Stack currently:

| Address | Value |
|---|---|
| 03FC | 55 |
| 03FD | 20 |
| 03FE | FF |
| 03FF | 22 |

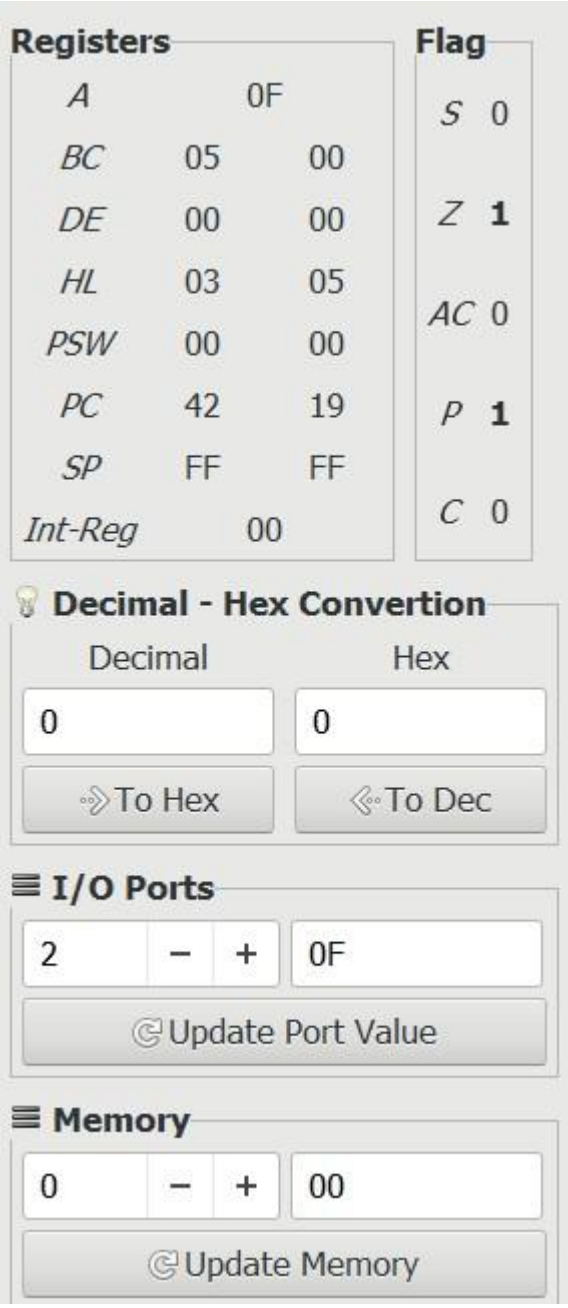
POP H:

- L ← 55 (from 03FC)
- H ← 20 (from 03FD)

HL = 2055H

| | |
|---|---|
| 8 | A multiplicand is stored in memory location XX50H and a multiplier is stored in location XX51H. Write a program to<br>1. Transfer the two numbers from memory loations to HL pair<br>2. Store a product at output port 02H<br>MVI A,05H<br>STA 2650H<br>MVI A,03H<br>STA 2651H<br><br>LHLD 2650H<br><br>MOV B,L<br>MOV C,H<br><br>MVI A,00H |

| | |
|---|---|
| | LOOP: ADD B<br>DCR C<br>JNZ LOOP<br><br>OUT 02H<br>HLT |

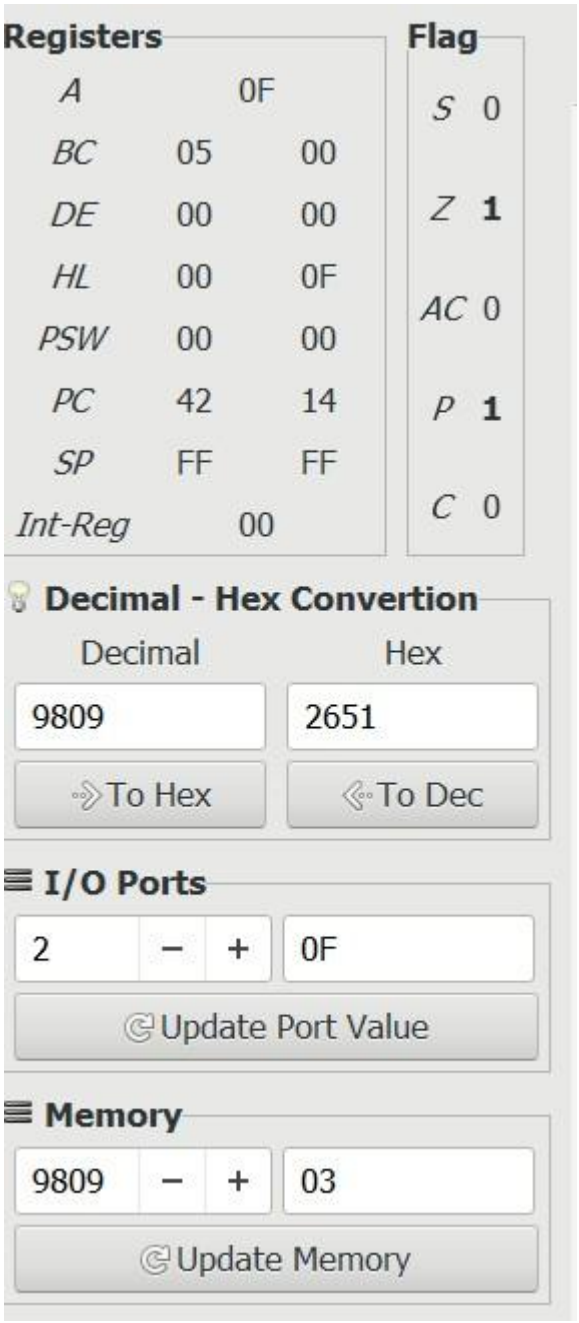



Write a subroutine to

1. Multiply numbers stored into HL pair
2. Return the result to HL pair

MVI A,05H

```
STA 2650H
MVI A,03H
STA 2651H

LHLD 2650H

CALL MUL

MOV A,L
OUT 02H
HLT

MUL: MOV B,L
     MOV C,H
     MVI A,00H

LOOP: ADD B
      DCR C
      JNZ LOOP

     MOV L,A
     MVI H,00H
     RET
```


Registers
A 0F
BC 05 00
DE 00 00
HL 00 0F
PSW 00 00
PC 42 14
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Decimal - Hex Convertion
Decimal
Hex
9809
2651
To Hex
To Dec
I/O Ports
2
0F
Update Port Value
Memory
9809
03
Update Memory

9

3. The following program has a delay subroutine located at location 2060H. Read the program and answer the questions given at the end of the program.

| Memory Locations | Mnemonics | |
|---|---|---|
| 2000 | LXI SP,20CDH | Main Program |
| 2003 | LXI H,0000H | |
| 2006 | MVI B,0FH | |
| 2008 | CALL 2060H | |
| 200B | OUT 01H | |
| | DCR B | |
| | CONTO | |
| 2060 | PUSH H | Delay Subroutine |
| 2061 | PUSH B | |
| | MVI B,05H | |
| | LXI H,COUNT | |
| | POP B | |
| | POP H | |
| | RET | |

a. When the execution of the CALL instruction located at 200BH–200AH is completed, list the contents stored at 20CCH and 20CBH, the contents of the program counter, and the contents of the stack pointer register.

b. List the stack locations and their contents after the execution of the instructions PUSH H and PUSH B in the subroutine.

c. List the contents of the stack pointer register after the execution of the instruction PUSH B located at 2061H.

d. List the contents of the stack pointer register after the execution of the instruction RET in the subroutine.

**(a) After CALL execution**
Initial:
SP = 20CDH
Return Address = 200BH
→ High = 20H, Low = 0BH
Execution:
SP → 20CCH → store 20H
SP → 20CBH → store 0BH
Answer:
(20CCH) = 20H
(20CBH) = 0BH
PC = 2060H
SP = 20CBH

**(b) After PUSH H and PUSH B**
**PUSH H (H=00, L=00)**
SP → 20CA → store H (00)
SP → 20C9 → store L (00)
**PUSH B (B=0F, C=?? assume 00)**
SP → 20C8 → store B (0F)
SP → 20C7 → store C (00)
Stack:

20CCH → 20H
20CBH → 0BH
20CAH → 00H
20C9H → 00H
20C8H → 0FH
20C7H → 00H

**(c) SP after PUSH B (line 2061H)**
SP = 20C7H

**(d) After RET** POP
sequence:

- PCL ← 0BH
- PCH ← 20H

SP moves:
20C7 → 20C8 → 20C9 → 20CA → 20CB → 20CC → 20CD Final
SP = 20CDH